



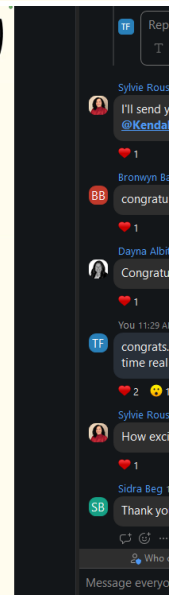
Foundations of Intervention Science: Setting the stage for scaling interventions

- Roughly 17 years to move Evidence -> Practice
 - Only about 14% of research efforts -> practice
- Low adoption and sustainment if evidence makes it to practice
 - Implementation cliff = interventions not used as intended to be used (that is, used with fidelity)
 - Impact cliff = evidence impact can decline over time
- Why the translation challenges?
 - Interventions don't readily (or appear to readily) meet the immediate needs of practitioners and their clientele or the public
 - Interventions are developed in ways that are challenging to implement by others (non-researchers) in routine practice



Foundations of Intervention Science: Setting the stage for scaling interventions

- Other Related Challenges
 - Reproducibility of findings = Obtaining consistent results from new data using similar methods and population
 - Up to 70% of other researchers failed to reproduce findings of others' studies; more than 50% unable to reproduce their own findings.
 - Low strength of findings
 - Insufficient documentation of details and necessary information and resources for others to replicate or apply the results
 - Generalizability/Representation: Relevance and evidence across diverse populations.
 - Does it work for my patients/clients/students?



i'm not trying to reproduce, thats the problem, if it's local why would it produce the same results. It's more than one way to skin a cat. of we looked at endstates and objective, and you cannot baseline. why are we trying to replicate processes, when we should aim to replicate results, recognizing it will always look different. 114 percent success is insanity. Each population has be addressed. Ecosystem, adaptation. Why are we trying to force a square peg in the round hole



Why does attention to intervention science matter for this topic of Scaling Interventions?

- Because the business of scaling is hard!
- Having a solid idea and grounding of an intervention in intervention science is useful in thinking about the scaling of the intervention.
 - Clear sense of the evidence for a given intervention can be used to guide decisions and policies to support uptake of the intervention
 - What works for whom, when, where and how?



What is this business of Scaling?

- Scaling or spread of interventions across settings, systems, communities and populations with goal of broader impact
 - Scaling Up: Spread to similar settings and consumers/clients/patient populations where and with whom the intervention has been tested
 - Scaling Out: Spread to different settings and/or populations
- "Scalability" of interventions – strength of the evidence, demand/need for the intervention, ability to readily implement the intervention or practice
 - What do we know about the evidence - and implementation - of the intervention to inform scaling?

What do we know here, and we are capturing but one intervention is not going to work in all places. The intervention should apply local

Intervention Development Science

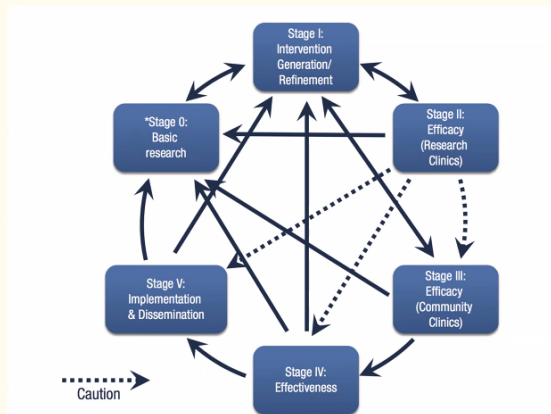
- What is a Behavioral Intervention?
 - Any intervention that uses behavioral, cognitive, affective or social processes to stimulate change in an individual's or group's behavior to promote health and well-being [and learning]. —*National Institute on Aging website*
- "The Thing" to be implemented

We shouldn't just implement a thing, we should implement the right thing, at the right location, at the right time, to solve the right problem. Expecting a 17 year old research, to behave against a real world issue. The piloting should be adaptable.

Behavioral Intervention Development Science: The NIH Stage Model



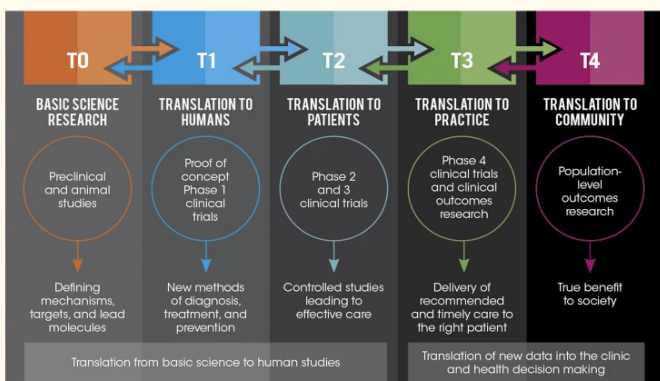
- Focus on mechanisms, how does the intervention work to change behavior?
- Opportunities to learn about relevant implementation-related contexts and outcomes at each phase of research



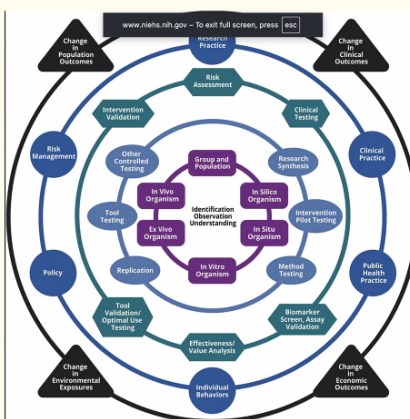
“Intervention development process is incomplete until an intervention is maximally efficacious **and** implementable with fidelity in community”

Onken LS, Carroll KM, Shoham V, Cuthbert BN, Riddle M. Reenvisioning clinical science: Unifying the discipline to improve the public health. *Clin Psychol Sci*. 2014;2(1):22-34. doi: 10.1177/21677702613407932

This doesn't work unless it's all local. You can bring it back and compare, but if you are trying to make every outcome the same at every location and through the process.



T Framework: National Center for Advancing Translational Sciences (NCATS)



Translational Research Framework: National Institute of Environmental Health Sciences

- ❖ **COMMON MISSION:** Guide a process to establish **Internal Validity** (does The Thing work under rigorously controlled conditions) as well as **External Validity** (does The Thing work in real world and what does it take to implement)